JUAN MIGUEL ZÚÑIGA-UMAÑA

May 27, 1991 @ juanmizuma@gmail.com +506 8859 9922 San Pedro Montes de Oca, San José, Costa Rica
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EXPERIENCE

Researcher
LANOTEC, CeNAT-CONARE
Aug 2023 – Ongoing Pavas, San José, Costa Rica

- Green synthesis of metallic nanoparticles for human health applications
- Analyzing ultrastructural patterns of Myxomycetes sporocarp for dispersion dynamics
- Automation of fungal pathogen spore ultrastructural patterns via ML
- Analyzing effects of naturally synthesized NPs on microbial growth

Research & Quality Assistant
LANOTEC, CeNAT-CONARE
2021 – 2023 San José, Costa Rica

- Research collaboration, lab management, reagent tracking, chemical regulatory liaison
- Knowledge transfer, scientific writing, experimental support & characterization analysis

Project: South-South Bilateral
Costa Rica – Uruguay
2021 – 2023

- Event logistics, audiovisual material, content and space preparation

Project: SEM Image Automation
2022 – Ongoing

- Coordination with students, labs, and interinstitutional researchers

Research Assistant
Plant Biotech Lab, School of Biology, UCR
2018 – 2020 San José, Costa Rica

- DNA extraction protocols from coffee leaf rust spores (*Hemileia vastatrix*)
- Genetic & ultrastructural diversity analysis of coffee leaf rust in Costa Rica
- Data analysis for laboratory-associated research

Statistical Services Unit Assistant
School of Statistics, UCR
2016 – 2017 San José, Costa Rica

- Information transfer, documentation review, data selection

Laboratory Assistant
Forest Resources Lab, School of Biosystems Eng., UCR
2014 – 2016 San José, Costa Rica

- Sample processing, Myxogastrid (slime mold) culture establishment
- Preliminary morphological identification of amoebozoans
- Training new collaborators; quotes & literature procurement

Laboratory Assistant
Plant Pathology Lab, School of Biology, UCR
2011 – 2013 San José, Costa Rica

- Macrofungi sample processing for microscopic identification

Collection Assistant
Luis Fornier O. Herbarium, School of Biology, UCR
2011 – 2013 San José, Costa Rica

- Collection maintenance

PROJECTS

Green Synthesis of AuNPs for glioblastome treatment
CeNAT, Pavas, San José, Costa Rica
2025-2027

SOFTSKILLS

Learning Potential | Team Work | Flexibility | Professionalism | Responsibility | Work Under Pressure | Agile | Proactive

PROFESSIONAL SKILLS

- R ●●●○○
- Python ●●○○○

STRENGTHS

- Lab Skills and Analytic instruments

DLS | TGA | UV-vis | FTIR | Holotomographic microscopy | NP green synthesis | DSC

- Development Tools – DB

R | GIS | Python | SAS

EDUCATION

B.Sc. in Biology
Faculty of Basic Sciences, Biology School, University of Costa Rica

MSc. in Bioinformatics and Systems Biology
Faculty of Medicine, University of Costa Rica
In progress 2025-

LANGUAGES

- English ●●●●○

- Green synthesis of metallic and biopolimeric nanoparticles, analytical techniques, biological activity / (Size/Charge/Shape controlled AuNPs obtained from the leaves extract of Acerola, modified to be be applied in radiotreatment of multiform glioblastome

Development of a biodegradable thermal foliar coating based on silica extracted from organic pineapple waste for thermal stress in tropical crops

2026-2027

- Evaluation of the performance of a bio-based film for foliar protection by radiative passive cooling, influence on plant metabolism and biophysical analysis

RECENT PUBLICATIONS

2026. Castillo-Henríquez, L., Agüero-Hidalgo, P., Zúñiga-Umaña, J.M., de Oca-Vásquez, G.M., Arce-Vásquez, F., Pereira-Vega, Z., Bahloul, B., Corvis, Y., & Vega-Baudrit, J.R. DoE-assisted green synthesis of AgNPs using Nephelium lappaceum peel extract: Size optimization enabling antibacterial and antioxidant activity.

Physchem, 6(2), 20.

https://doi.org/10.3390/physchem6020020

2026. Berkelmann, D., Zúñiga-Umaña, J.M., Chaverri, P., Solano, W., & Gatica-Arias, A. Fungal diversity associated with coffee leaf rust (Hemileia vastatrix) pustules based on ITS1 amplicon sequencing.

World J Microbiol Biotechnol, 42, 153.

https://doi.org/10.1007/s11274-026-04905-1

2025. Chacón-Calderón, A., Zúñiga-Umaña, J.M., Villarreal, C., Vega-Baudrit, J.R., Pereira-Reyes, R., & Corrales-Ureña, Y.R. Mangrove soil as a natural catalyst for green synthesis of silver nanoparticles.

Front Chem, 13, 1589836.

https://doi.org/10.3389/fchem.2025.1589836

2025. Zúñiga-Umaña, J.M., Paniagua, S.A., Brenes, R.C., & Vega-Baudrit, J.R. Microplastic pollution in Costa Rican marine ecosystems: Origins, ecotoxicological impacts, and mitigation strategies.

Mar Policy, 179, 106772.

https://doi.org/10.1016/j.marpol.2025.106772

2025. Díaz-Canales, P., Zúñiga-Umaña, J.M., Rojas Camacho, P., & Rojas, C. Myxomycetes recorded in rapid assessments in Central America and Mexico during 2012-2019.

Slime Molds, 5, V5A10.

https://doi.org/10.5281/zenodo.14974841

2024. Hanley, G., Vargas Jiménez, S., Baltodano Viales, E., Zúñiga-Umaña, J.M., Vega-Baudrit, J., Murillo Rodríguez, Y., & Castillo Henríquez, L. Quality-by-design driven approach in the formulation of an anti-ulcer and gastroprotective oral suspension.

Drug Dev Ind Pharm, 50(7), 646–657.

https://doi.org/10.1080/03639045.2024.2383932